Quantum 2 HW2

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Technion

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1 Question 1

1.1 Part 1

What are the probabilities $P_\pm$ to measure a magnetic moment $\mu_z = \pm\mu_0$, when the neutron's position doesn't matter?

Since the neutron's position doesn't matter, the probability of finding a specific magnetic moment will be the sum of the probabilities in all positions, therefore

$$P_\pm = \int_{\mathbb{R}^3} \mathrm{d}P = \int_{\mathbb{R}^3} |\psi_\pm(\boldsymbol{r},t)|^2 \,\mathrm{d}\boldsymbol{r} \tag{1}$$

1.2 Part 2

Calculate $\langle \mu_x \rangle$ in the state $|\psi(\boldsymbol{r},t)\rangle$ in terms of $\psi_\pm(\boldsymbol{r},t)$.

First we notice that $\mu_x = \mu_0\sigma_x$, therefore

$$\begin{aligned}
\langle \mu_x \rangle &= \mu_0 \langle \psi(t)| \sigma_x |\psi(t)\rangle & (2)\\
&= \mu_0 \int_{\mathbb{R}^3} \psi^*(\boldsymbol{r},t)\sigma_x\psi(\boldsymbol{r},t)\,\mathrm{d}\boldsymbol{r} & (3)\\
&= \mu_0 \int_{\mathbb{R}^3} \left(\psi_-^*(\boldsymbol{r},t)\langle -| + \psi_+^*(\boldsymbol{r},t)\langle +|\right)\sigma_x\left(\psi_-(\boldsymbol{r},t)|-\rangle + \psi_+(\boldsymbol{r},t)|+\rangle\right)\mathrm{d}\boldsymbol{r} & (4)\\
&= \mu_0 \int_{\mathbb{R}^3} \left(\psi_-^*(\boldsymbol{r},t)\langle -| + \psi_+^*(\boldsymbol{r},t)\langle +|\right)\left(|+\rangle\langle -| + |-\rangle\langle +|\right)\left(\psi_-(\boldsymbol{r},t)|-\rangle + \psi_+(\boldsymbol{r},t)|+\rangle\right)\mathrm{d}\boldsymbol{r} & (5)\\
&= \mu_0 \int_{\mathbb{R}^3} \psi_+^*(\boldsymbol{r},t)\psi_-(\boldsymbol{r},t) + \psi_-^*(\boldsymbol{r},t)\psi_+(\boldsymbol{r},t)\,\mathrm{d}\boldsymbol{r} & (6)\\
&= 2\mu_0 \int_{\mathbb{R}^3} \Re\left(\psi_-^*(\boldsymbol{r},t)\psi_+(\boldsymbol{r},t)\right)\mathrm{d}\boldsymbol{r} & (7)
\end{aligned}$$

1.3 Part 3

Calculate $\langle \boldsymbol{r} \rangle$ and $\langle \boldsymbol{p} \rangle$ in the state $|\psi(t)\rangle$.

$$
\begin{aligned}
\langle \boldsymbol{r} \rangle &= \langle \psi(t) | \, \boldsymbol{r} \, | \psi(t) \rangle && (8)\\
&= \int \psi^*(\boldsymbol{r},t) \boldsymbol{r} \psi(\boldsymbol{r},t) \, \mathrm{d}\boldsymbol{r} && (9)\\
&= \int \left(\psi_-^*(\boldsymbol{r},t) \langle - | + \psi_+^*(\boldsymbol{r},t) \langle + | \right) \boldsymbol{r} \left(\psi_-(\boldsymbol{r},t) | - \rangle + \psi_+(\boldsymbol{r},t) | + \rangle \right) \mathrm{d}\boldsymbol{r} && (10)\\
&= \int \left(|\psi_-(\boldsymbol{r},t)|^2 + |\psi_+(\boldsymbol{r},t)|^2 \right) \boldsymbol{r} \, \mathrm{d}\boldsymbol{r} && (11)
\end{aligned}
$$

$$
\begin{aligned}
\langle \boldsymbol{p} \rangle &= \langle \psi(t) | \, \boldsymbol{p} \, | \psi(t) \rangle && (12)\\
&= -i\hbar \int \psi^*(\boldsymbol{r},t) \boldsymbol{\nabla} \psi(\boldsymbol{r},t) \, \mathrm{d}\boldsymbol{r} && (13)\\
&= -i\hbar \int \left(\psi_-^*(\boldsymbol{r},t) \langle - | + \psi_+^*(\boldsymbol{r},t) \langle + | \right) \nabla \left(\psi_-(\boldsymbol{r},t) | - \rangle + \psi_+(\boldsymbol{r},t) | + \rangle \right) \mathrm{d}\boldsymbol{r} && (14)\\
&= -i\hbar \int \psi_-^*(\boldsymbol{r},t) \nabla \psi_-(\boldsymbol{r},t) + \psi_+^*(\boldsymbol{r},t) \nabla \psi_+(\boldsymbol{r},t) \, \mathrm{d}\boldsymbol{r} && (15)
\end{aligned}
$$

1.4 Part 4

Assume that $c_\pm(t) = \alpha_\pm$ and is not time dependent. We'll define $\psi'_+(\boldsymbol{r},t) = \psi'_-(\boldsymbol{r},t) \equiv \psi(\boldsymbol{r},t)$ so that the neutron's state now is

$$|\psi(\boldsymbol{r},t)\rangle = \psi(\boldsymbol{r},t) \begin{pmatrix} \alpha_+ \\ \alpha_- \end{pmatrix}$$

How do your answers to parts 2 and 3 change?

$$
\begin{aligned}
\langle \mu_x \rangle &= 2\mu_0 \int\limits_{\mathbb{R}^3} \Re \left(\psi_-^*(\boldsymbol{r},t) \psi_+(\boldsymbol{r},t) \right) \mathrm{d}\boldsymbol{r} && (16)\\
&= 2\mu_0 \int\limits_{\mathbb{R}^3} \Re \left(\psi^*(\boldsymbol{r},t) \psi(\boldsymbol{r},t) \alpha_-^* \alpha_+ \right) \mathrm{d}\boldsymbol{r} && (17)\\
&= 2\mu_0 \Re \left(\alpha_-^* \alpha_+ \right) && (18)
\end{aligned}
$$

$$\langle \boldsymbol{r} \rangle = \int \left(|\psi_-(\boldsymbol{r},t)|^2 + |\psi(\boldsymbol{r},t)|^2\right) \boldsymbol{r} \, \mathrm{d}\boldsymbol{r} \tag{19}$$

$$= \int \left(\left(|\alpha_-|^2 + |\alpha_+|^2\right) |\psi(\boldsymbol{r},t)|^2\right) \boldsymbol{r} \, \mathrm{d}\boldsymbol{r} \tag{20}$$

$$= \int |\psi(\boldsymbol{r},t)|^2 \boldsymbol{r} \, \mathrm{d}\boldsymbol{r} \tag{21}$$

$$\langle \boldsymbol{p} \rangle = -i\hbar \int \psi_-^*(\boldsymbol{r},t)\nabla\psi_-(\boldsymbol{r},t) + \psi_+^*(\boldsymbol{r},t)\nabla\psi_+(\boldsymbol{r},t) \, \mathrm{d}\boldsymbol{r} \tag{22}$$

$$= -i\hbar \int \left(|\alpha_-|^2 + |\alpha_-|^2\right) \psi^*(\boldsymbol{r},t)\nabla\psi(\boldsymbol{r},t) \, \mathrm{d}\boldsymbol{r} \tag{23}$$

$$= -i\hbar \int \psi^*(\boldsymbol{r},t)\nabla\psi(\boldsymbol{r},t) \, \mathrm{d}\boldsymbol{r} \tag{24}$$

2 Question 2

1. Write down the Hamiltonian for the Neutron. Write down the time-dependent Schrödinger equation for the state of the Neutron and show it can be decomposed into two similar equations for $\psi_\pm$
2. Show that if $|\psi(t)\rangle$ is of the form in the previous question then

$$\frac{d}{dt}\left(\int |\psi_\pm(\boldsymbol{r},t)|\,\mathrm{d}\boldsymbol{r}\right) = 0$$

   What can we deduce for the probability to measure $\mu_z = \pm\mu_0$
3. Long question I won't copy
4. Use Ehrenfest's theorem to find $\frac{d}{dt}\langle\boldsymbol{r}\rangle_\pm$ and $\frac{d}{dt}\langle\boldsymbol{p}\rangle_\pm$. Solve the resulting system of equations and find the time development for both. Give a physical interpretation for the result and explain why the original beam splits into two beams.
5. What is the distance D between the two marks on the screen when $L = 1m$? Assume $L \gg D$. What is the ratio of the lengths D and the Neutron's radius?

2.1 Hamiltonian

As we know the Hamiltonian is

$$H = \frac{p^2}{2m} - \boldsymbol{\mu}\cdot\boldsymbol{B} = \frac{p^2}{2m}\mathbb{1} - \mu_0\left(B_0 + bz\right)\sigma_z \tag{25}$$

And the time-dependent Schrödinger equation is

$$i\hbar\frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle \tag{26}$$

If we replace the generic wave equation with the vector wave equation from question 1 then the ODE that is the Schrödinger equation becomes two:

$$i\hbar\frac{d}{dt}\psi(\boldsymbol{r},t)\begin{pmatrix}\alpha_+\\ \alpha_-\end{pmatrix} = H_\pm|\psi(t)\rangle = -\frac{\hbar^2}{2m}\nabla^2\psi_\pm \mp \mu_0\left(B_0 + bz\right)\psi_\pm \tag{27}$$

The $\mp$ appearing because of the inverted magnetic moment.

2.2 Integral Derivative

Since $|\psi(t)\rangle$ is of the form $|\psi(\boldsymbol{r},t)\rangle = \psi(\boldsymbol{r},t)\begin{pmatrix}\alpha_+\\ \alpha_-\end{pmatrix}$, then

$$\int |\psi_\pm(\boldsymbol{r},t)|\,\mathrm{d}\boldsymbol{r} = |\alpha_\pm|^2 \tag{28}$$

then it is a constant and therefore also not time-dependent, which gives us the needed identity. This expression is also the same as the probability of measuring $\mu_{\pm}$, which means the probabilities of those are also constant.

2.3 Part 3

From the structure of the integral we can tell that this is some kind of observation value. The values noted represent the probability of finding the particle in a certain position.

2.4 Time Development

Recalling Ehrenfest's Theorem:

$$i\hbar\frac{d}{dt}\langle\boldsymbol{\mu}\rangle = \langle[H,\boldsymbol{\mu}]\rangle \tag{29}$$

then (shortening $\boldsymbol{p}_{\pm}$ to p)

$$\begin{aligned}
\frac{d}{dt}\langle\boldsymbol{r}_{\pm}\rangle &= -\frac{i}{\hbar}\langle[H,\boldsymbol{r}]\rangle &(30)\\
&= -\frac{i}{\hbar}\int\psi^*[H,\boldsymbol{r}]\psi\,\mathrm{d}\boldsymbol{r} &(31)\\
&= -\frac{i}{\hbar}\int\psi^*[\frac{p^2}{2m},\boldsymbol{r}]\psi\,\mathrm{d}\boldsymbol{r} &(32)\\
&= -\frac{i}{2m\hbar}\int\psi^*[p^2,\boldsymbol{r}]\psi\,\mathrm{d}\boldsymbol{r} &(33)
\end{aligned}$$

where

$$\begin{aligned}
[p^2,r] &= p^2r - rp^2 &(34)\\
&= ppr - rpp &(35)\\
&= p(rp-(rp-pr)) - (pr-(pr-rp))p &(36)\\
&= p(rp-[r,p]) - (pr-[p,r])p &(37)\\
&= prp - p[r,p] - prp + [p,r]p &(38)\\
&= 2i\hbar p &(39)
\end{aligned}$$

therefore

$$\begin{aligned}
-\frac{i}{2m\hbar}\int\psi^*[p^2,\boldsymbol{r}]\psi\,\mathrm{d}\boldsymbol{r} &= \frac{1}{m}\int\psi^*p\psi\,\mathrm{d}\boldsymbol{r} &(40)\\
&= \frac{\langle\boldsymbol{p}_{\pm}\rangle}{m} &(41)
\end{aligned}$$

and

$$\frac{d}{dt}\langle \boldsymbol{p}_\pm \rangle = -\frac{i}{\hbar}\langle [H, \boldsymbol{p}_\pm] \rangle \tag{42}$$

$$\tag{43}$$

we notice immediately that

$$\frac{d}{dt}\langle p_x \rangle = \frac{d}{dt}\langle p_y \rangle = 0 \tag{44}$$

and now

$$\begin{aligned}
\frac{d}{dt}\langle p_{z\pm} \rangle &= -\frac{i}{\hbar}\int \psi^*[H, p_{z\pm}]\psi \, \mathrm{d}\boldsymbol{r} && (45)\\
&= -\frac{i}{\hbar}\int \psi^*[-\mu_0 bz, p_{z\pm}]\psi \, \mathrm{d}\boldsymbol{r} && (46)\\
&= \frac{i\mu_0 b}{\hbar}\int \psi^*[z, p_{z\pm}]\psi \, \mathrm{d}\boldsymbol{r} && (47)\\
&= - \mp \mu_0 b \int \psi^* \psi \, \mathrm{d}\boldsymbol{r} && (48)\\
&= \pm \mu_0 b && (49)
\end{aligned}$$

Therefore

$$\boldsymbol{p}_\pm = \begin{pmatrix} p_0 \\ 0 \\ \pm\mu_0 bt \end{pmatrix} \qquad \boldsymbol{r} = \begin{pmatrix} 0 \\ 0 \\ \frac{\pm\mu_0 bt^2}{2m} \end{pmatrix} \tag{50}$$

2.5 Finding D

$$D = \langle z_+ \rangle - \langle z_- \rangle = \frac{\mu_0 bt^2}{m} = \frac{\mu_0 bL^2}{mv^2} = \frac{\mu_0 bL^2}{2E} \tag{51}$$

$$= \text{[inserting the numbers into a calculator]} = \tag{52}$$

3 Question 3

We'll look at the following time-dependent hamiltonian:

$$H(t) = \frac{(P+mv)^2}{2m} + \frac{1}{2}m\omega^2 (X-vt)^2 - \frac{1}{2}mv^2$$

where X, P are canonically conjugate operators with $[X, P] = i\hbar\mathbb{1}$ and $m, v, \omega \in \mathbb{R}$.

1. Find a time-independent hamiltonian using the operator $S(t) = e^{\frac{i}{\hbar}vtP}$.
2. Explain the physical meaning to the operator $S(t)$ and of the result.

3.1 Eliminating Time-Dependence

As we know, given an operator $S(t)$ (not necessarily the one given in the question) and a time-dependent hamiltonian H, we can get a time-independent hamiltonian by

$$H_S(t) = i\hbar\frac{dS}{dt}S^\dagger(t) + S(t)H(t)S^\dagger(t) \tag{53}$$

We'll check if the operator we were given gives us the required result

$$\begin{aligned} i\hbar\frac{dS}{dt}S^\dagger(t) &= i\hbar\cdot\frac{i}{\hbar}vPS(t)S^\dagger(t) &(54)\\ &= -vP &(55)\end{aligned}$$

$$S(t)H(t)S^\dagger(t) = S(t)\left(\frac{(P+mv)^2}{2m} + \frac{1}{2}m\omega^2(X-vt)^2 - \frac{1}{2}mv^2\right)S^\dagger(t) \tag{56}$$

We can immediately see that the first and last elements will be zero (S is a function of P and thus commutates with P, the last element is a constant) therefore

$$\begin{aligned} S(t)H(t)S^\dagger(t) &= S(t)\left(\frac{1}{2}m\omega^2(X-vt)^2\right)S^\dagger(t) &(57)\\ &= S(t)\left(\frac{1}{2}m\omega^2\left(X^2-2vtX+v^2t^2\right)\right)S^\dagger(t) &(58)\\ &= \frac{1}{2}m\omega^2\left(S(t)\left(X^2-2vtX+v^2t^2\right)S^\dagger(t)\right) &(59)\\ &= \frac{1}{2}m\omega^2\left(\left(X^2-2vtX+v^2t^2\right)S(t)S^\dagger(t) - [X^2-2vtX+v^2t^2, S(t)]S^\dagger(t)\right) &(60)\\ &= \frac{1}{2}m\omega^2\left(\left(X^2-2vtX+v^2t^2\right) - [X^2-2vtX+v^2t^2, S(t)]S^\dagger(t)\right) &(61)\end{aligned}$$

To continue we calculate:

$$
\begin{aligned}
[X, S] &= \left[X, \exp\left(\frac{i}{\hbar} vtP\right)\right] && (62)\\
&= [X, P] \cdot \frac{i}{\hbar} vt \exp\left(\frac{i}{\hbar} vtP\right) && (63)\\
&= -vtS(t) && (64)
\end{aligned}
$$

and

$$
\begin{aligned}
\left[X^2, S\right] &= X\left[X, S\right] + \left[X, S\right] X && (65)\\
&= -vt\left(XS(t) + S(t)X\right) && (66)\\
&= -vt\left(2XS(t) + [S, X]\right) && (67)\\
&= -vt\left(2XS(t) + vtS(t)\right) && (68)\\
&= -2vtXS(t) - v^2t^2S(t) && (69)
\end{aligned}
$$

therefore

$$
\begin{aligned}
S(t)H(t)S^\dagger(t) &= \frac{1}{2} m\omega^2 \left(\left(X^2 - 2vtX + v^2t^2\right) - [X^2 - 2vtX + v^2t^2, S(t)]S^\dagger(t)\right) && (70)\\
&= \frac{1}{2} m\omega^2 \left(\left(X^2 - 2vtX + v^2t^2\right) - \left(-2vtXS(t) + v^2t^2S(t)\right) S^\dagger(t)\right) && (71)\\
&= \frac{1}{2} m\omega^2 X^2 && (72)
\end{aligned}
$$

Which is indeed time-independent.

3.2 Physical Meaning

The hamiltonian looked like it is of a harmonic oscillator which is moving at a certain velocity too. The operator S brought the hamiltonian back to the form of a stationary harmonic oscillator. This likely means the operator S is a galileo transformation operator. The result means that we can solve the hamiltonian for the system in its own rest frame when possible (at least classically, special relativity is probably more complex)

4 Question 4

1. What is the hamiltonian to 0-th order, and how will the states $|g\rangle, |e\rangle$ develop>?
2. What will the states of $|g\rangle, |e\rangle$ be after a half-pi pulse?
3. For an initial state $|g\rangle$, after a half-pi pulse, we'll turn the electric field off and let the system develop in time. What is the state now?
4. After the free propagation we'll send another half-pi pulse. What will the state be after this? What is the likelihood to find the atom in the excited state ($P_e(\Delta)$)?
5. We can measure $P_e(\Delta)$ and tune ω using it. The sensitivity is defined by $S \equiv \left|\frac{dP_e}{d\Delta}\right|$.
   (a) What is the meaning of this value?
   (b) What is our clock's sensitivity?
   (c) When will we get maximum sensitivity?
   (d) What is our error range?
6. Estimate how long until the clock will be 1 second off.

4.1 0-th Order Hamiltonian

Given that $\Omega \gg \Delta$, The 0th order hamiltonian is just the element that doesn't include Δ

$$H = \frac{1}{2}\hbar\Omega\sigma_x \tag{73}$$

and to find the time development of the states, since the hamiltonian is time indepdendent

$$\begin{aligned} U(t) = e^{-\frac{i}{\hbar}Ht} = e^{-\frac{1}{2}i\Omega\sigma_x} && (74)\\ = \cos\left(\frac{\Omega t}{2}\right)\mathbb{1} - i\sin\left(\frac{\Omega t}{2}\right)\sigma_x && (75)\\ = \cos\left(\frac{\Omega t}{2}\right)(|g\rangle\langle g| + |e\rangle\langle e|) - i\sin\left(\frac{\Omega t}{2}\right)(|g\rangle\langle e| + |e\rangle\langle g|) && (76) \end{aligned}$$

Therefore

$$\begin{aligned} |g(t)\rangle &= U(t)|g\rangle && (77)\\ &= \cos\left(\frac{\Omega t}{2}\right)|g\rangle - i\sin\left(\frac{\Omega t}{2}\right)|e\rangle && (78)\\ |e(t)\rangle &= U(t)|e\rangle && (79)\\ &= \cos\left(\frac{\Omega t}{2}\right)|e\rangle - i\sin\left(\frac{\Omega t}{2}\right)|g\rangle && (80) \end{aligned}$$

4.2 After Half-Pi Pulse

The states will be

$$
\begin{aligned}
|g'\rangle &= \cos\left(\frac{\pi}{4}\right)|g\rangle - i\sin\left(\frac{\pi}{4}\right)|e\rangle && (81)\\
&= \frac{1}{\sqrt{2}}|g\rangle - \frac{i}{\sqrt{2}}|e\rangle && (82)\\
|e'\rangle &= \cos\left(\frac{\pi}{4}\right)|e\rangle - i\sin\left(\frac{\pi}{4}\right)|g\rangle && (83)\\
&= \frac{1}{\sqrt{2}}|e\rangle - \frac{i}{\sqrt{2}}|g\rangle && (84)
\end{aligned}
$$

4.3 Free Development

When the electric field is turned off, the hamiltonian is

$$H = \frac{1}{2}\hbar\Delta\sigma_z \tag{85}$$

so the time development operator is

$$
\begin{aligned}
U_2(t) &= e^{-\frac{i}{\hbar}Ht} = e^{-\frac{1}{2}i\Delta\sigma_z} && (86)\\
&= \cos\left(\frac{\Delta t}{2}\right)\mathbb{1} + i\sin\left(\frac{\Delta t}{2}\right)\sigma_z && (87)
\end{aligned}
$$

therefore

$$
\begin{aligned}
|g''\rangle &= U_2(t)|g'\rangle && (88)\\
&= \cos\left(\frac{\Delta T}{2}\right)|g'\rangle + i\sin\left(\frac{\Delta T}{2}\right)(|e\rangle\langle e| - |g\rangle\langle g|)|e'\rangle && (89)\\
&= \cos\left(\frac{\Delta T}{2}\right)|g'\rangle + i\sin\left(\frac{\Delta T}{2}\right)(|e\rangle\langle e| - |g\rangle\langle g|)\left(\frac{1}{\sqrt{2}}|e\rangle - \frac{i}{\sqrt{2}}|g\rangle\right) && (90)\\
&= \cos\left(\frac{\Delta T}{2}\right)|g'\rangle + i\sin\left(\frac{\Delta T}{2}\right)\left(\frac{1}{\sqrt{2}}|e\rangle + \frac{i}{\sqrt{2}}|g\rangle\right) && (91)\\
&= \frac{1}{\sqrt{2}}\left(\cos\left(\frac{\Delta T}{2}\right)(|g\rangle - |e\rangle) + i\sin\left(\frac{\Delta T}{2}\right)(|e\rangle + |g\rangle)\right) && (92)\\
&= \frac{1}{\sqrt{2}}\left(\left(\cos\left(\frac{\Delta T}{2}\right) + i\sin\left(\frac{\Delta T}{2}\right)\right)|g\rangle - \left(\cos\left(\frac{\Delta T}{2}\right) - i\sin\left(\frac{\Delta T}{2}\right)\right)|e\rangle\right) && (93)\\
&= \frac{1}{\sqrt{2}}\left(e^{\frac{i\Delta t}{2}}|g\rangle - ie^{-\frac{i\Delta t}{2}}|e\rangle\right) && (94)
\end{aligned}
$$

4.4 Another Pulse

We are back to using the electric field time propagator, therefore

$$
\begin{aligned}
|g'''\rangle &= U(\frac{\pi}{2\Omega})\,|g''\rangle && (95)\\
&= \left(\cos\left(\frac{\pi}{4}\right)(|g\rangle\langle g| + |e\rangle\langle e|) - i\sin\left(\frac{\pi}{4}\right)(|g\rangle\langle e| + |e\rangle\langle g|)\right)\frac{1}{\sqrt{2}}\left(e^{\frac{i\Delta t}{2}}|g\rangle - ie^{-\frac{i\Delta t}{2}}|e\rangle\right) && (96)\\
&= \frac{1}{\sqrt{2}}\left(\cos\left(\frac{\pi}{4}\right)\left(e^{\frac{i\Delta t}{2}}|g\rangle - ie^{-\frac{i\Delta t}{2}}|e\rangle\right) - i\sin\left(\frac{\pi}{4}\right)\left(e^{\frac{i\Delta t}{2}}|e\rangle - ie^{-\frac{i\Delta t}{2}}|g\rangle\right)\right) && (97)\\
&= \frac{1}{2}\left(\left(e^{\frac{i\Delta t}{2}} - e^{-\frac{i\Delta t}{2}}\right)|g\rangle - i\left(e^{\frac{i\Delta t}{2}} + e^{-\frac{i\Delta t}{2}}\right)|e\rangle\right) && (98)\\
&= \sinh\left(\frac{i\Delta t}{2}\right)|g\rangle - i\cosh\left(\frac{i\Delta t}{2}\right)|e\rangle && (99)\\
&= i\sin\left(\frac{\Delta t}{2}\right)|g\rangle - i\cos\left(\frac{\Delta t}{2}\right)|e\rangle && (100)
\end{aligned}
$$

therefore

$$
\begin{aligned}
P_e(\Delta) &= |\langle e|g'''\rangle|^2 && (101)\\
&= \cos^2\left(\frac{\Delta t}{2}\right) && (102)
\end{aligned}
$$

4.5 Tuning and Sensitivity

4.5.1 Meaning

The meaning of the sensitivity S is how strongly a deviation from the resonant frequency affects the operation of the clock. A high value means even small deviations will cause significant changes to the probability of the states flipping.

4.5.2 Our Clock's Sensitivity

$$
S = \left|\frac{dP_e(\Delta)}{d\Delta}\right| = \left| = t\sin\left(\frac{\Delta t}{2}\right)\cos\left(\frac{\Delta t}{2}\right)\right| = \left|\frac{t}{2}\sin(\Delta t)\right| \tag{103}
$$

4.5.3 Maximum Sensitivity

We can see the maximums will be found at $t = \frac{(2k+1)\pi}{2\Delta}$

4.5.4 Error Range

$$
\delta\Delta = \frac{3\pi - \pi}{2T} = \frac{\pi}{T} \tag{104}
$$

4.6 Time Until 1s Error

Using the given equation

$$\delta t = \frac{\delta f}{f_0} t \implies t = \frac{f_0 \delta t}{\delta f} \tag{105}$$

therefore

$$t = \frac{9\,192\,631\,770\,\mathrm{Hz} \cdot 1\,\mathrm{s}}{\pi\,\mathrm{s}^{-1}} = 92.72\,\mathrm{yr} \tag{106}$$